

VLSI Physical Design Fresher Handbook

Physical Design Flow

Netlist -> Floorplan -> Power Plan -> Placement -> CTS -> Routing -> Signoff.

Important interview discussion points and practical understanding required.

Floorplanning

Die Area, Core Area, Utilization, Aspect Ratio, Macro Placement, Halos, Channels.

Important interview discussion points and practical understanding required.

Power Planning

Power Rings, Stripes, Meshes, IR Drop, Electromigration, Decap Cells.

Important interview discussion points and practical understanding required.

Placement

Global Placement, Legalization, Optimization, Congestion Reduction.

Important interview discussion points and practical understanding required.

CTS

Clock Tree Synthesis balances clock paths and minimizes skew.

Important interview discussion points and practical understanding required.

CTS Terms

Skew, Latency, Jitter, Insertion Delay, Uncertainty, Useful Skew.

Important interview discussion points and practical understanding required.

Clock Gating

Reduces dynamic power by disabling unnecessary clock switching.

Important interview discussion points and practical understanding required.

Routing

Global Routing and Detailed Routing connect all nets.

Important interview discussion points and practical understanding required.

STA

Static Timing Analysis checks setup and hold timing.

Important interview discussion points and practical understanding required.

Setup & Hold

Setup: data before clock edge. Hold: data after clock edge.

Important interview discussion points and practical understanding required.

IR Drop

Voltage drop due to power network resistance.

Important interview discussion points and practical understanding required.

Electromigration

Metal degradation due to high current density.

Important interview discussion points and practical understanding required.

Tap Cells

Connect n-well to VDD and substrate to GND.

Important interview discussion points and practical understanding required.

End Cap Cells

Protect row boundaries.

Important interview discussion points and practical understanding required.

Decap Cells

Reduce voltage fluctuations.

Important interview discussion points and practical understanding required.

Libraries

.lib, LEF, DEF, SPEF, Verilog.

Important interview discussion points and practical understanding required.

Linux

pwd, ls, cd, mkdir, rm, cp, mv, grep, find.

Important interview discussion points and practical understanding required.

TCL

Variables, loops, conditions, procedures.

Important interview discussion points and practical understanding required.

Interview Questions

Utilization, congestion, CTS, setup, hold, IR drop, EM, routing.

Important interview discussion points and practical understanding required.

Top Fresher Interview Q&A;

Q: What is utilization?

A: $\text{Standard Cell Area} / \text{Core Area} \times 100$.

Q: Ideal utilization?

A: Typically 65%–75%.

Q: What is congestion?

A: Routing demand exceeds available routing resources.

Q: Why CTS?

A: To distribute clock with controlled skew and latency.

Q: What is setup violation?

A: Data arrives late.

Q: What is hold violation?

A: Data arrives too early.

Q: What is IR Drop?

A: Voltage drop in power network.

Q: What is EM?

A: Metal wear-out due to current density.